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CASE REPORT

Chronic occupational *N*, *N*-dimethylformamide poisoning induced death: a case report

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Abstract *N*, *N*-dimethylformamide (DMF) is a toxic solvent that is widely used in many industries. Death directly attributed to DMF poisoning is rare. We report the case of a 40-year-old female who died of liver failure after occupational exposure to DMF poisoning over the course of 70 days, followed by 64 days treatment in hospital. Based on clinical procedures, autopsy findings, and environmental air quality assessment, the death in this case was confirmed to be the result of chronic DMF poisoning. This case gives further insight into occupational chronic poisoning-induced death, especially with negative toxicological analysis of rapid metabolism toxins poisoning.

Keywords Forensic toxicology · Forensic pathology · *N*, *N*-dimethylformamide (DMF) · Occupational disease · Negative toxicological analysis

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Introduction

N, *N*-dimethylformamide (DMF), a solvent and raw material of organic compounds, was first synthesized in 1960 and has been widely used in fiber, leather, dye, and chemical industries. DMF is toxic to humans and affects multiple organs following exposure via inhalation, ingestion, or skin contact [1]. When DMF enters into the body, it is metabolized in the liver by mixed-function oxidase to become monomethyl and *N*-hydroxymethyl-*N*-methylformamide (HMMF) [2]. HMMF undergoes thermolytic degradation to form *N*-methylformamide (NMF) and formaldehyde. The pathway for the metabolism of NMF is oxidation of the formyl group, resulting in *N*-acetyl-*S*-(*N*-methylcarbamoyl) cysteine (AMCC), which binds to DNA, RNA, and various proteins and causes liver and kidney injury [3].

The toxic effects of DMF exposure in humans are varied and can include skin eruption, gastric irritation, and hepatotoxicity [4, 5]. Liver injury appears to be the main hallmark of DMF poisoning [6]. The severity of liver injury after exposure to DMF for humans is directly related to the concentration of DMF. The lethal dose (LD_{50}) of DMF in rats is 2800 ± 1200 mg/kg via oral administration and 5000 ± 1430 mg/m^3 via acute inhalation [7]. The current DMF limit for occupational exposure in Europe and the United States is 30 mg/m^3 [8]. An exposure level below 10 mg/m^3 is unlikely to induce severe hepatic dysfunction, but DMF concentrations between 10 and 30 mg/m^3 may cause liver injury in sensitive populations [9].

DMF poisoning is a frequently encountered occupational phenomenon. However, reports of death due to chronic DMF poisoning are very rare. Here, we report a case of 40-year-old female who died of liver failure due to chronic DMF poisoning.

Case report

A 40-year-old woman working in a Polyurethanes/Polyvinyl Chloride (PU/PVC) factory for 70 days described a progressive decline in health, including poor appetite, fatigue, nausea, vomiting, and jaundice, which were ignored by the patient for more than 2 months. The patient was not admitted to the hospital until she developed severe jaundice, anasarca, and hepatalgia. Abdomen CT scans and type-B ultrasonic tests revealed uneven liver density, hepatic atrophy, thickening of the gallbladder wall, an enlarged spleen, and peritoneal effusion. Laboratory tests showed elevated levels of increased total and direct bilirubin (216.2 and 104.5 μmol/L, respectively), alanine aminotransferase (848U/L), aspartate aminotransferase (963 U/L), blood urea nitrogen (22.1 mmol/L), and creatinine (234.4 μmol/L). A tentative diagnosis of viral hepatitis was made, despite negative findings for hepatitis surface antigens and antibodies. Treatment to confer anti-inflammation and hepatoprotection were initiated, during which time liver damage progressed as evidenced by clinical manifestations and liver function tests. Measures were then taken to maintain liver function, including blood transfusions and placement of the patient on a waiting list for a liver transplantation. Despite aggressive treatment in the hospital for a total of 64 days, the woman died.

The Disease Control Center performed an air quality assessment in the occupational environment of the decedent. The data from 12 sampling points showed DMF concentrations ranging from 41.4 to 131 mg/m^3; all sampling points exceeded the exposure limit of $\leq$40 mg/m^3 set by the Chinese government. Investigators later discovered that the factory had been instructed to suspend production due to air DMF concentrations prior to employment of the decedent.

Results of a forensic autopsy revealed significant jaundice of the skin and sclera. Internal examination revealed that both the surface and the cut surface of all organs (particularly the liver) were jaundiced. The liver was found to be soft, with significant atrophy (weight, 610 g) (Fig. 1). The gallbladder was swollen and filled with bile (70 mL). Splenic atrophy (weight 80 g) and cortical thickening of the kidney (0.6 cm) were observed. The brain showed severe encephaledema. There was a peritoneal effusion (2000 mL), and bilateral pleural effusions of 1900 mL were noted.

Microscopic evaluation of the liver tissue showed collapse of liver lobules in which massive hepatic cell necrosis and proliferation of fiber tissue were observed. The portal area revealed ductal proliferation, cholestasis, and infiltrating lymphocytes (Fig. 2). The renal tubular epithelium

Fig. 1 Hepatic atrophy. Both the external surface and the cut surface of the liver were *yellow*; the liver had atrophied significantly

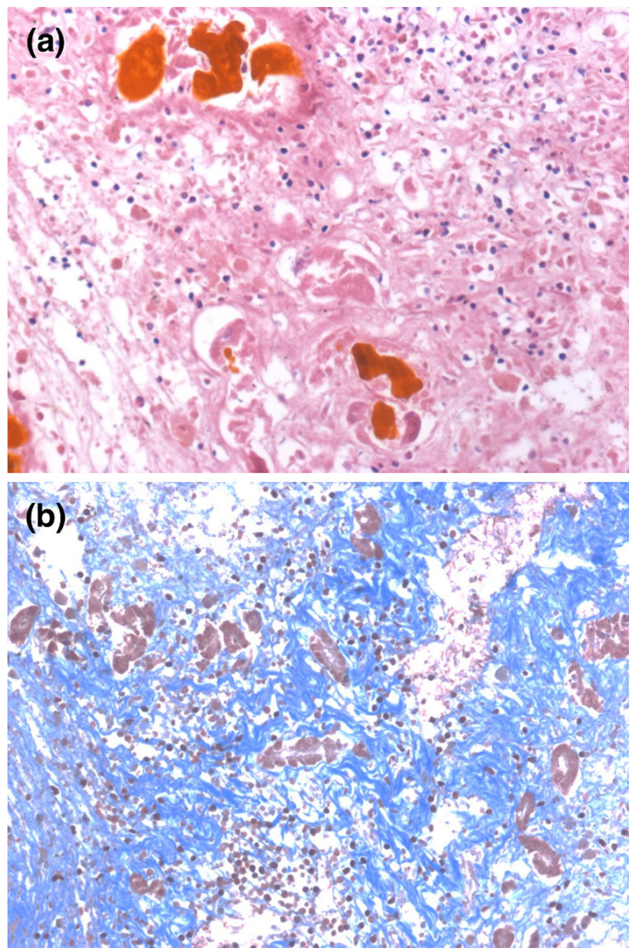

Fig. 2 Hepatic necrosis. **a** Massive hepatic cell necrosis, the hyperplasia of the bile duct accompanied with biliary cholestasis, and scattered inflammatory cell infiltration (HE stain, ×200). **b** A large section of fibrous tissue hyperplasia (Masson stain, ×200)

was swollen and several bile casts were seen within the renal tubule lumen.

The cause of death based on the clinical history, laboratory testing, and autopsy findings was documented as severe hepatic necrosis resulting from DMF poisoning.

Discussion

We searched NCBI PubMed, Google Scholar, and the China Knowledge Resource Integrated Database for all years and all languages using the key words "DMF poisoning AND (death OR autopsy)" and performed a retrospective analysis of 6 reports of death caused by DMF poisoning. The results are summarized in Table 1 [10–14]. Based on our research, all reported cases of DMF poisoning-induced death occurred in specific occupations, including leather and PU/PVC industries. Clinical symptoms of DMF-poisoned decedents revealed liver injury and jaundice as the most common features. Fatigue, nausea, hydroabdomen, gastrointestinal bleeding, and disseminated intravascular coagulation (DIC) were primary symptoms, and liver failure was listed as cause of death in all cases.

For forensic pathologists, evidence of DMF poisoning remains challenging. This is because presence of DMF in the blood or urine is difficult to detect due to the rapid metabolism of this agent [15]. It has been reported that an average DMF concentration of 2.8 mg/L in the blood becomes undetectable at 4 h postexposure [16]. Santoni et al. [17] demonstrated that DMF is only detectable in blood or urine after acute exposure to higher concentrations. In reviewed cases, the toxicological analyses of blood and urine samples were all negative for DMF. The long treatment period is regarded as a contributing factor to the negative toxicology results in these cases. Therefore, we recommend that blood and/or urine samples should be collected and stored as early as possible in cases of unexplained hepatic necrosis for more accurate toxicological analysis. In addition, Osunsanya et al. [18] determined that NMF was present in urine of individuals exposed to DMF, and theorized that the metabolites of DMF might be present in liver tissue. Thus, the detection of NMF in the liver

Table 1 Cases of DMF poison induced death [10–14]

Authors	Publication year	Age/sex	Work place	Concentration of DMF in work place (mg/m^3)	Days of exposure	Days from treatment to death	Clinical symptoms	DMF analysis in blood/urine	Mechanism of death
Wang et al.	2006	18/M	Leather factory	102.52	71	13	Liver dysfunction; jaundice; nausea; vomiting; UGB; DIC	Negative/negative	Hepatotoxicity
Gu	2009	23/M	PU/PVC factory	104.7	85	48	Liver dysfunction; fatigue, jaundice; UGB; DIC	Negative/negative	Hepatotoxicitywith gastric perforation
Hong et al.	2009	40/F	PU/PVC factory	204.7	/	45	Liver dysfunction; hepatic atrophy; hydroabdomen; splenauxe.	Negative/negative	Hepatotoxicity
Ding et al.	2011	38/F	PU/PVC factory	54.3	18	/	Liver dysfunction; hepatic atrophy; hydroabdomen	Negative/negative	Hepatotoxicity
Tong et al.	2014	23/F	Leather factory	157.1	60	15	Liver dysfunction; fatigue; hydroabdomen; jaundice	Negative/negative	Hepatotoxicity with gastric perforation
Current case	/	40/F	PU/PVC factory	131	70	64	Liver dysfunction; nausea; vomiting; fatigue; jaundice; poor appetite; severe encephaledema; hepatic atrophy	Negative/negative	Hepatotoxicity

UGB upper gastrointestinal bleeding

might be of significance when exploring a diagnosis of DMF poisoning.

As it is currently understood, the pathological changes in liver or kidney are not specific for the diagnosis of DMF poisoning. However, Redlich et al. [19] reported different pathological changes for acute versus chronic DMF poisoning. In this study, biopsy results showed massive hepatic necrosis and microvesicular fatty degeneration in chronic DMF poisoning cases, while focal hepatocellular necrosis and macrovesicular fatty degeneration were observed in cases of acute DMF poisoning. Furthermore, some studies have found pathological changes in kidney tissue including swollen renal tubular epithelial cells and few bile casts of mice and rabbits [20]. Pathological lesions of hepatotoxity and nephrotoxicity were also observed in our case.

A diagnosis of virus hepatitis and/or other causes of liver injury should be excluded through a comprehensive analysis of the decedents medical history. The patient in this case was reported to be healthy prior to working in the factory. In addition, the deceased had no history of medication use or excessive alcohol intake that would contribute to injury of the liver or other organs. Importantly, an environmental air assessment and physical evaluation of co-workers can provide evidence for a diagnosis of DMF poisoning. In this case report, DMF concentrations in the deceased work environment exceeded the legal limits for occupational exposure (a crucial factor in determining the cause of death), and several colleagues of the deceased also suffered varying degrees of liver dysfunction. Therefore, based on Chinese standards for occupational disease, the present case is in accordance with chronic occupational DMF poisoning.

In summary, the current case can serve as a reference for the diagnosis of cases of rapid metabolism toxins poisoning based on medical records, autopsy findings, and environmental exposures. Moreover, this case also highlights the importance of raising awareness among medical staff, especially occupational health physicians, to ensure that liver injury due to occupational exposure is recognized as early as possible.

Key points

1. Chronic occupational DMF poisoning can result in liver failure and death.
2. The cause of severe hepatic necrosis due to DMF poisoning should be carefully evaluated by excluding other possible causes.
3. Clinical procedure analysis, autopsy findings, and environmental air quality assessment were comprehensively carried out to confirm that the death was caused by chronic DMF poisoning in this case.
4. This case gives further insight into occupational chronic poisoning-induced deaths, especially those associated with negative toxicological analysis of rapid metabolism toxins poisoning.

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References

1. Potter HP. Toxicity of dimethylformamide. Lancet. 1974;2:1084.
2. Twiner MJ, Hirst M, Valenciano A, Zacharewski TR, Dixon SJ. *N*, *N*-dimethylformamide modulates acid extrusion from murine hepatoma cells. Toxicol Appl Pharmacol. 1998;153:143–51.
3. Van den Bulcke M, Rosseel MT, Wijnants P, Buylaert W, Belpaire FM. Metabolism and hepatotoxicity of *N*, *N*-dimethylformamide, *N*-hydroxymethyl-*N*-methylformamide, and *N*-methylformamide in the rat. Arch Toxicol. 1994;68:291–5.
4. Potter HP. Dimethylformamide-induced abdominal pain and liver injury. Arch Environ Health. 1973;27:340–1.
5. Hamada M, Abe M, Tokumoto Y, et al. Occupational liver injury due to *N*, *N*-dimethylformamide in the synthetics industry. Intern Med. 2009;48:1647–50.
6. Cheng J, Liu Q, Xu B, et al. Effect of *N*, *N*-dimethylformamide on oxidation or antioxidation status among occupational exposed workers. Chinese J Prev Med. 2014;48:28–32.
7. Zhang Y. Manual of Toxic Chemicals Toxicity. Joint publishing house of peking medical college and beijing union medical college; 1993.
8. Randall B. Disposition of toxic drugs and chemicals in man. 7th ed. Ireland: Elsevier; 2004.
9. Nomiyama T, Uehara M, Miyauchi H, Imamiya S, Tanaka S, Seki Y. Causal relationship between a case of severe hepatic dysfunction and low exposure concentrations of *N*, *N*-dimethylformamide in the synthetics industry. Ind Health. 2001;39:33–6.
10. Wang M, Xie X. Analysis on a case of occupational dimethylformamide poisoning death. J Ind Med. 2006;20:272–3.
11. Gu Y. Occupational dimethylformamide poisoning death: a case report. J Occup Heal Damage. 2009;24:181–2.
12. Hong Y, Ma Y. Occupational dimethylformamide-induced severe poisoning death: a case report. Chiese Occup Med. 2009;36:143–4.
13. Ding Y, Ying G, Jin L. A case of occupational dimethylformamide poisoning death. Occup Heal Emerg Rescue. 2011;29(2):107–8.
14. Tong Z, Shi J, Zhu X, Zhu B. Death after exposure to dimethylformamide: a case report. Chinese J Ind Hyg Occup Dis. 2014;32:285–6.
15. Mraz J. Gas chromatographic method for the determination of *N*-acetyl-*S*-(*N*-methylcarbamoyl)cysteine, a metabolite of *N*, *N*-dimethylformamide and *N*-methylformamide, in human urine. J Chromatogr. 1988;431:361–8.
16. Kimmerle G, Eben A. Metabolism studies of *N*,*N*-dimethylformamide. II. Studies in persons. Int Arch Arbeitsmed. 1975;34:127–36.
17. Santoni G, Bavazzano P, Perico A, et al. High-performance liquid chromatographic determination of *N*-methylformamide and *N*-

methyl-*N*-(hydroxymethyl)-formamide in human urine. J Chromatogr. 1992;581:287–92.
18. Osunsanya T, Adejoro B, King B. Biological monitoring of workers exposed to dimethylformamide in a textile polyurethane unit. Occup Med (Lond). 2001;51:374–9.
19. Redlich CA, West AB, Fleming L, True LD, Cullen MR, Riely CA. Clinical and pathological characteristics of hepatotoxicity associated with occupational exposure to dimethylformamide. Gastroenterology. 1990;99:748–57.
20. Qiu Q, Miao X, Lu Z, Hu G. Experimental study on acute dimethylformamide poisoning and detoxicification in mice and rabbits. Zhejiang Med J. 2006;28:349–52.